Craftax runs compared: achievements and what unlocks them

family/next: 1.1M env steps each, size50m, 30 fixed worlds (tools/death_eval.py). next_lam0 is masked (reads the game's action rules): counted as cheating. next_nomask, the honest run, has no result yet (its first attempt was killed at ~74k steps). Updated 2026-11-08.

Every number is the mean episode-final achievement count over 30 evaluation episodes, ± its standard error over those worlds; unlock rates are the share of episodes in which an achievement fired; return is the mean raw Craftax return as a % of the maximum (226). Every checkpoint is scored on the same fixed worlds (env.reseed, audit B05), so comparisons are paired world by world. The error bars cover the worlds only, not training seeds.

Plain DreamerV3 (abl_plain, seed 0) scores 4.77 ± 0.28, level with vanilla DreamerV3 on the original code (4.63 ± 0.22); normalized return 1.71% and 1.65%.

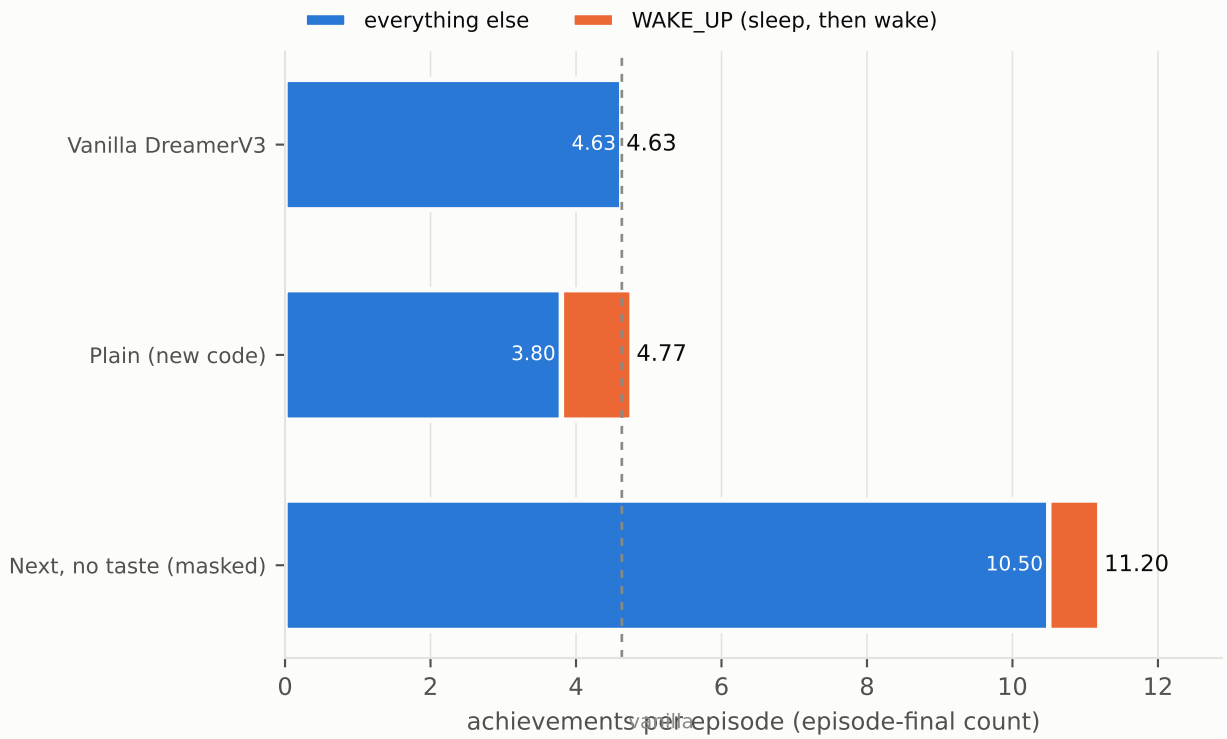
The next agent with the rule mask and taste on the manager at lam_mgr 0 (no taste), next_lam0, scores 11.20 ± 0.41 (4.56% return) on these worlds and 11.30 ± 0.27 on 60 worlds, the best run of the project. It is MASKED: the mask reads the game's action rules, so it is counted as cheating and is not a headline result. The honest run, next_nomask (the same agent with the mask off), was killed at ~74k env steps by an out-of-memory crash of the box on 2026-11-08 and has to be rerun; it is the run that shows whether the next agent is good without cheating.

run	ach. ± SE	w/o wake	life	return	table	pickaxe	sword	stone	drink
Vanilla DreamerV3	4.63 ± 0.22	4.63	295	1.65%	63%	23%	3%	0%	77%
Plain (new code)	4.77 ± 0.28	3.80	265	1.71%	50%	3%	10%	0%	73%
Next, no taste (masked)	11.20 ± 0.41	10.50	284	4.56%	100%	100%	83%	97%	77%

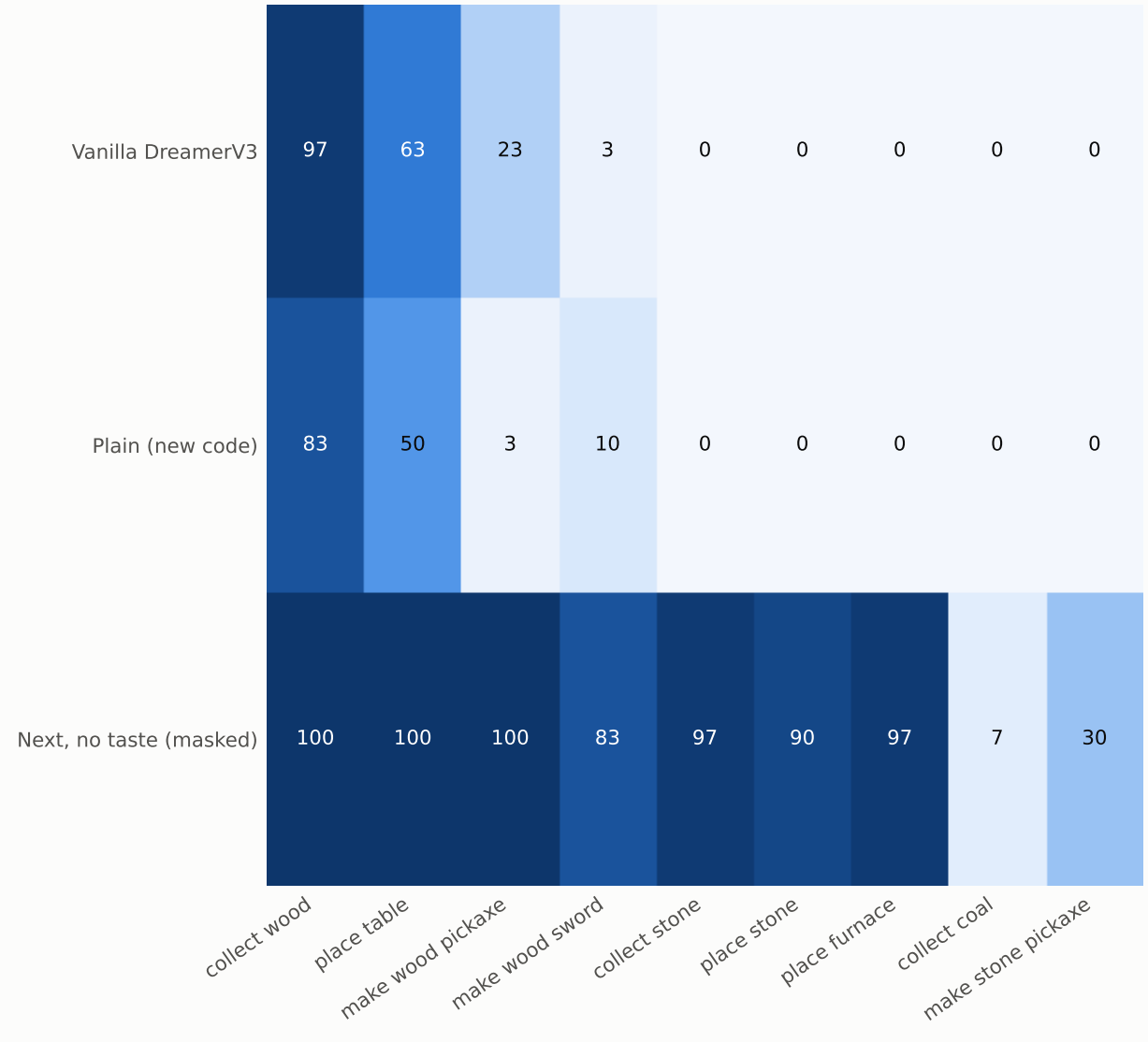
ach. ± SE = mean episode-final achievements ± standard error over the 30 worlds (one training seed, so seed variation is not included). w/o wake = the same minus WAKE_UP. life = mean lifespan in steps. return = raw Craftax return, % of 226. Rates are % of episodes. Vanilla DreamerV3: the original checkpoint (code before the 2026-10 fixes), re-scored on the fixed worlds. Plain (new code): `--configs craftax size50m --env.craftax.survival none`; the row shows seed 0 (seed 1 also scores 4.77). Next, no taste: next_lam0, the next agent (RSSM-2 memory, curiosity, 13 named goals, shared critic, hold 32) with the rule mask and a taste that adds exactly zero (lam_mgr 0); MASKED, counted as cheating. A 30-world score moves by about ±0.55 between world sets: gaps under ~1.1 between single runs are not evidence.

Achievements per run, and the early tech tree

Mean episode-final achievements per run, split into WAKE_UP and the rest; unlock rate of the early tech-tree rungs



Unlock rate (% of episodes), early tech tree in dependency order



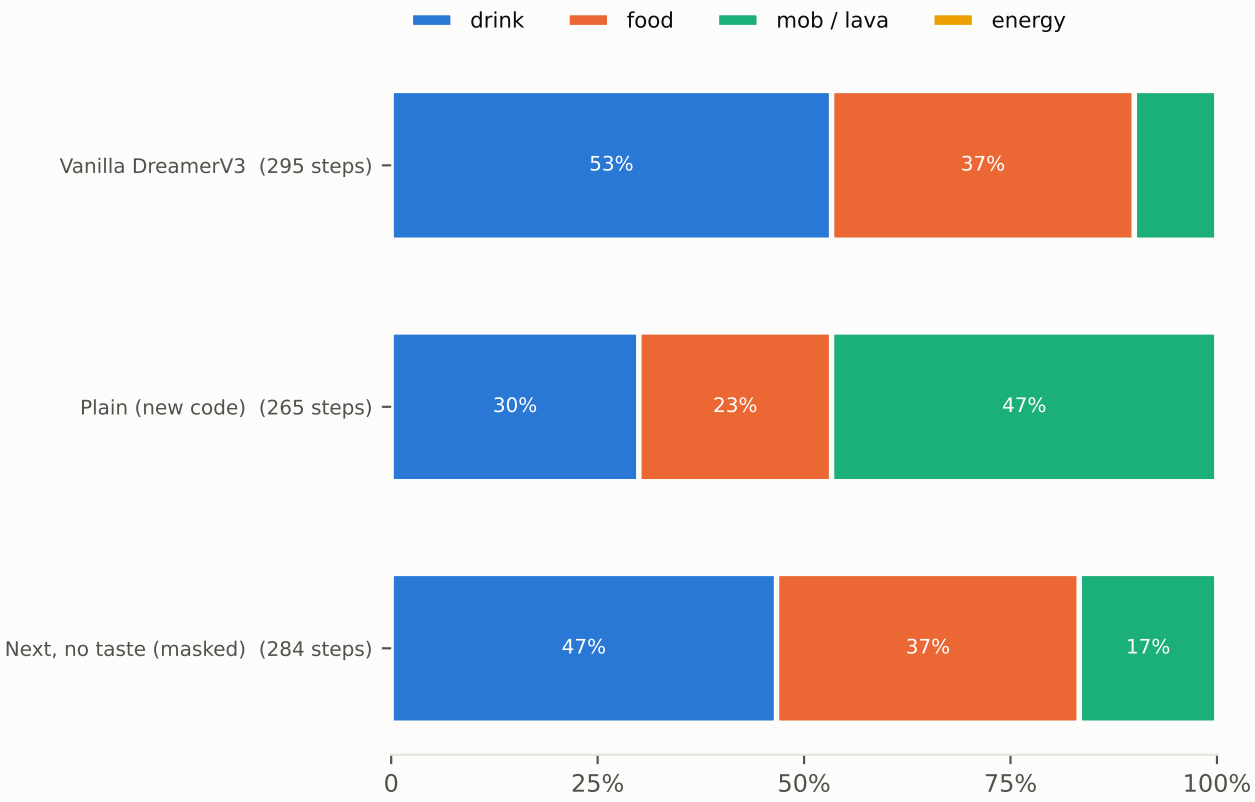
Every achievement any run unlocked

% of evaluation episodes in which each achievement fired. The other 50 of 67 were never unlocked by any run.



Cause of death and lifespan

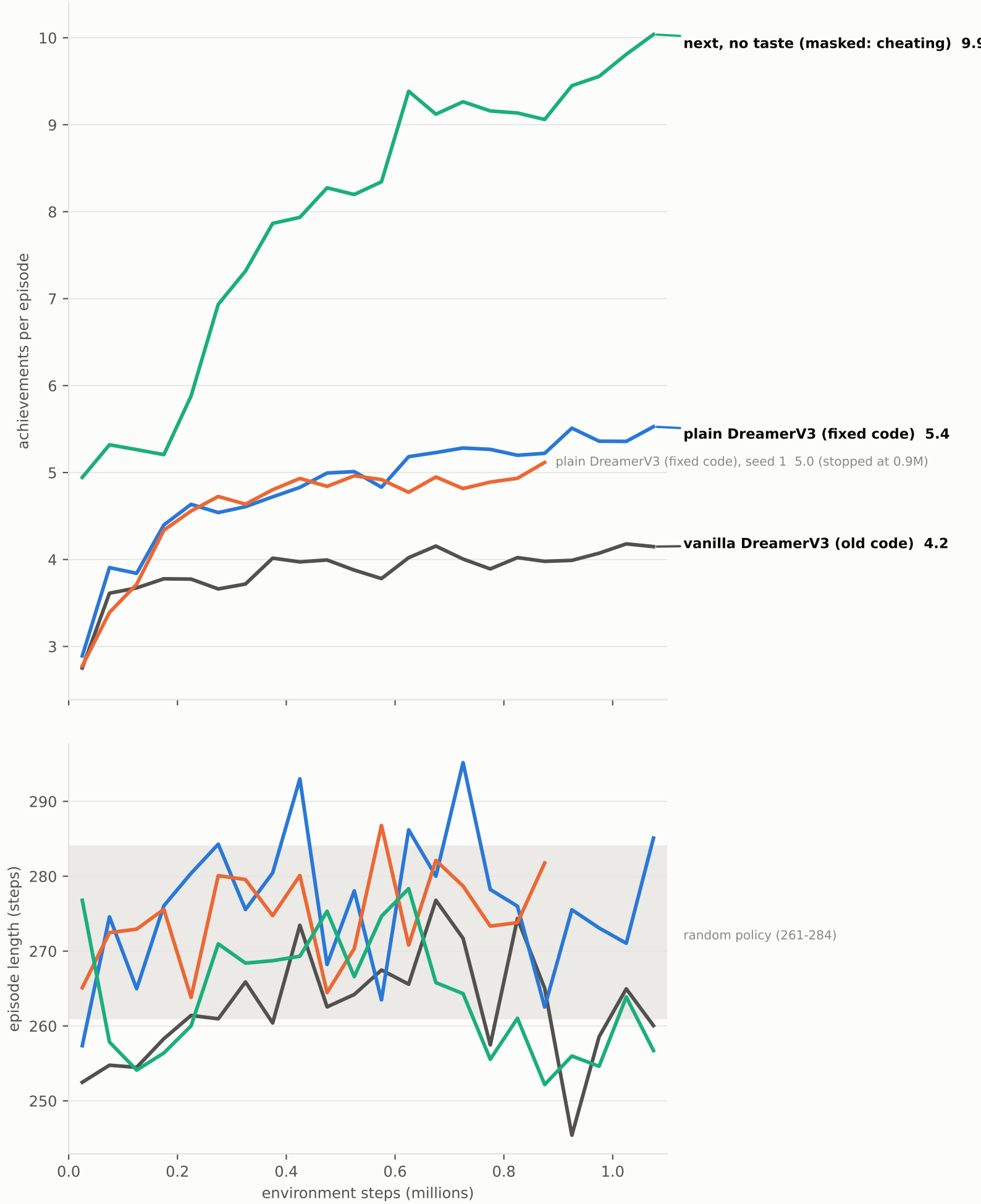
Cause of death (share of evaluation episodes) and lifespan; and, if the results list one, how much a run moves between its own checkpoints



Random policy lives 261-284 steps; doing nothing lives 333.

Training curves: every run

Rebuilt from each run's replay; 50k-step bins; value = mean of the last 100k steps.



What the results say, and what would settle the rest

The plain baseline

The fixed code does not move plain DreamerV3: 4.77 against 4.63, well inside the noise of a 30-world score (about ±0.55 between world sets; a gap under ~1.1 between single runs is not evidence). Both stay at the wood tier: no stone is collected in any episode, and lifespans (265-295 steps) sit near the random-policy band (261-284).

The next agent with the rule mask is far above the baseline

next_lam0 scores 11.20 ± 0.41 against plain's 4.77 ± 0.28, and 11.30 ± 0.27 on 60 worlds. It makes the stone sword in 50% and the stone pickaxe in 30% of episodes on worlds 0-29, which no plain run does. But the rule mask reads the game's action rules, so this run is counted as cheating and does not rank against honest agents. Without the mask the same family scored 6.03 ± 0.33 for manager v1.2 (v12_nomask, family/v) and 5.77 for RSSM-2 plus the manager without curiosity (abl_manager); both are on other branches.

What would settle the rest

next_nomask (the next agent without the rule mask) has no result: its first attempt (started 2026-11-08 09:26 CST) was killed at ~74k env steps by an out-of-memory crash of the box at 11:35, and has to be rerun. Score it on 60 fixed worlds and compare with next_lam0 (11.30) and v12_nomask (6.03): that is the honest measure of the next agent. next_notaste crashed three times on GPU faults (Xid 69, 13) and has no result; next_lam0 stands in for it, since taste at lam_mgr 0 adds exactly zero. Taste on the manager (next_taste, family/taste) scores 9.80 ± 0.42 against next_lam0's 11.20 on the same worlds, so taste hurt; see docs/taste-on-manager.md.